

Robotic Systems I: 2025-2026 Projects

Instructor: Konstantinos Chatzilygeroudis (costashatz@upatras.gr)

PhD students: Konstantinos Asimakopoulos (asimakopoulos@ac.upatras.gr),

Lampros Printzios (lampros@ac.upatras.gr)

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Project 1: Escape Room Robot – Exploration, Mapping, and Key Discovery

Robot: Elephant Robotics MyAGV 2023 Pi omnidirectional mobile

Deployment: Real robot

In this project, teams will develop an autonomous robotic system that explores an unknown environment, identifies key elements, and uses them to unlock access to new areas. The robot starts in an unexplored environment containing doors, obstacles, and special objects acting as “keys”. Some doors may only be considered passable once the corresponding key has been detected.

The robot will be deployed in an unknown, obstacle-filled environment. Certain regions of the maze may be blocked by doors, represented by ArUco markers. A door is considered closed until the agent discovers its corresponding key, which is represented by a different unique ArUco marker. The mapping between each key-door pair (i.e., their associated ArUco IDs) is assumed to be known in advance.

The goal is to design a complete autonomy pipeline for exploration, perception, state estimation, and decision making. Teams must build or extend a mapping and localization system, detect landmarks, plan exploration actions, and decide which regions to visit next.

Core tasks.

- Explore an initially unknown environment using the onboard Lidar.
- Build and maintain a map of free space, obstacles, doors, and keys.
- Detect and localize keys and doors using the onboard camera.
- Decide when and how to revisit doors after finding relevant keys.
- Complete the full environmental traversal and return to the starting location.

Relevant methods. SLAM, frontier-based exploration, state estimation, object detection, data association, motion planning, task-level decision making.

Evaluation. Teams will be evaluated based on two criteria:

1. **Exploration completion** — how effectively the robot explores and maps the environment.
2. **Navigation performance after exploration** — once the map has been created, the robot must be able to navigate to any requested point in the maze. During testing, teams will be evaluated on the efficiency of the generated paths.

Project 2: Catch It on the Belt – Visual Tracking and Interception with a Robot Arm

Robot: Interbotix ViperX-300 6DOF robotic arm

Deployment: Real robot

In this project, teams will control a robotic arm tasked with detecting, tracking, intercepting, and grasping objects moving along a conveyor belt. Students are responsible not only for the robot arm control strategy, but also for developing the complete visual perception pipeline used to estimate the objects' pose and velocity.

The conveyor belt will move objects with varying speeds and initial poses. Importantly, teams will have no direct control over, or feedback from, the conveyor system itself. Instead, they must infer the motion of the objects entirely through visual observations from onboard or external cameras.

Using these observations, teams must estimate object trajectories, predict suitable interception points, and command the robotic arm to grasp the relevant objects at the correct time. Successfully grasped objects must then be placed at a specified target location.

Core tasks.

- Detect the moving object using RGB or RGB-D perception.
- Estimate object position and velocity online.
- Predict the future object pose on the belt.
- Plan an interception motion for the arm.
- Execute the motion while respecting joint limits and timing constraints.

Relevant methods. Visual tracking, Kalman filtering, trajectory prediction, inverse kinematics, SE(3) pose control, constrained optimization, motion timing.

Evaluation. Teams will be evaluated on catch success rate, interception accuracy, robustness to different belt speeds, tracking quality, timing precision, and smoothness of the robot motion.

Project 3: Perception-Guided Arm Control in Unstructured Scenes

Robot: Elephant Robotics MyArm 300 Pi 7DOF robotic arm

Deployment: Real robot

In this project, teams will develop a robotic manipulation system that combines visual perception and control to interact with objects in an unstructured environment. Rather than executing predefined motions, the robot must continuously use visual feedback to adapt its behavior to changing object poses, partial occlusions, scene clutter, and sensing uncertainty.

A camera mounted in the workspace will observe a box containing multiple objects. The robot’s task is to identify and manipulate only a specific subset of target objects (for example, locating and grasping 5 rubber ducks from a box containing 20 mixed items). Not all target objects may be visible initially, requiring the system to reason about partially occluded scenes and adapt as objects move during interaction.

Possible subtasks include object detection and recognition, visual servoing, grasp planning, object alignment, and pick-and-place operations in cluttered environments. Teams must design both the perception pipeline and the robot control strategy required to reliably identify, grasp, and relocate the desired objects.

Core tasks.

- Detect and localize task-relevant objects using RGB or RGB-D sensing.
- Estimate object pose relative to the robot.
- Generate arm motions based on the perceived scene.
- Adapt online to changes or errors in perception.
- Avoid obstacles and respect robot constraints.

Relevant methods. RGB-D perception, camera calibration, pose estimation, visual servoing, inverse kinematics, optimization-based control, obstacle avoidance, and SE(3) transformations.

Evaluation. Teams will be evaluated on task success rate, perception accuracy, control accuracy, robustness to object pose variation, collision avoidance, and ability to recover from perception errors.

Project 4: Bimanual Manipulation – Simulation - Only Coordinated Control with Simulated Forces

Robot: Franka Robotics Franka Emika Research 3 7DOF robotic arm (2 of them)

Deployment: MuJoCo simulation environment

In this project, teams will develop a bimanual manipulation system in which two robot arms must cooperatively transport a shared tray while maintaining the stability of an object placed on top. The task focuses on coordinated manipulation, closed-chain control, and reasoning about interaction forces between multiple manipulators and a shared payload.

The tray will be intentionally asymmetric, causing its center of mass to be unevenly distributed. In addition, a spherical object will be placed on top of the tray. Because the sphere can freely roll, even small orientation errors or sudden accelerations may destabilize the payload. Teams must therefore coordinate both robot arms to maintain smooth and stable tray motion throughout the task.

Core tasks.

- Model the asymmetric tray, the spherical payload, and the dual-arm grasp configuration.
- Generate coordinated bimanual motions for lifting, transporting, and placing the tray.
- Maintain relative pose and orientation constraints between the two end-effectors to ensure stable cooperative manipulation.
- Minimize tray tilt, oscillations, and sudden accelerations that could cause the sphere to roll or fall.
- In simulation, reason about and regulate contact forces, load sharing, and internal forces between the two manipulators.
- Transfer the resulting strategy to position-controlled real robot hardware.

Relevant methods. Bimanual inverse kinematics, constrained optimization, coordinated trajectory generation in $SE(3)$, grasp and closed-chain constraints, cooperative manipulation, load sharing, internal force regulation, stability-aware motion planning, and sim-to-real transfer.

Evaluation. Teams will be evaluated based on: successful completion of the transport task, stability of the tray during motion, ability to keep the spherical payload on the tray, smoothness and synchronization of the two robot arms, quality of the simulated force-aware behavior, and robustness to variations in tray mass distribution and initial sphere placement.

Project 5: Biped Locomotion Challenge – Simulation - Only Walking Control

Robot: Unitree Robotics G1 humanoid robot or Pal Robotics Talos humanoid robot

Deployment: MuJoCo simulation environment

In this project, teams will develop a simulated bipedal walking controller capable of generating stable locomotion through dynamic balance and center-of-mass control. The project is simulation-only and focuses on the fundamental principles of humanoid walking, including balance maintenance, weight shifting, and stable foot-to-foot transitions.

The core objective is to enable the humanoid robot to walk by continuously moving its center of mass (CoM) over the supporting foot during each step. Teams must design controllers that coordinate the robot’s posture, balance, and stepping behavior so that stable single-support phases can be achieved without falling.

Core tasks.

- Generate stable walking motions for a simulated humanoid robot.
- Plan and control center-of-mass (CoM) trajectories to achieve stable locomotion.
- Shift the robot’s weight appropriately to stabilize over the supporting foot during single-support phases.
- Coordinate foot placement and body motion to enable continuous stepping and balance maintenance.
- Maintain stability during transitions between double-support and single-support phases.
- Recover from small disturbances, balance errors, or imperfect foot placements during walking.

Relevant methods. Finite-state machines, center-of-mass planning, zero-moment point (ZMP) control, linear inverted pendulum models (LIPM), capture point methods, model predictive control, inverse kinematics, whole-body control, trajectory optimization, and stability analysis.

Evaluation. Teams will be evaluated based on their ability to achieve stable continuous walking, successfully stabilize over a single support foot, and maintain balance during support transitions. Additional evaluation criteria include walking distance and duration before falling, smoothness and naturalness of the generated gait, robustness to disturbances or modeling errors, and performance on unseen walking scenarios or terrain variations (e.g. small stairs).